

SPD

27 + 30 = 57

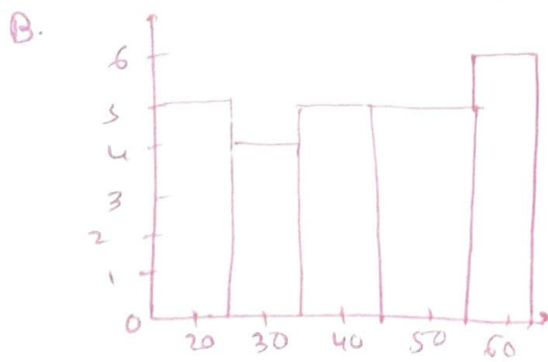
SOMAIYA
VIDYAVIHAR UNIVERSITY

(08) ab.

Semester: January 2026 – April 2026		
Maximum Marks: 30	Examination: In-Semester Examination	Duration : 1 hr.15 min.
Programme code: 03(UG)		
Programme: B.Tech Electronics and Telecommunication Engineering	Class: SY	Semester IV (SVU 2020/SVU 2023)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: EXTC
Course Code: 216h27C401	Name of the Course: Advanced Data Analytics (Honours Programme in Data Science for Emerging Analytics)	

Question No.		Max. Marks
Q1	The time taken (in seconds) by 25 students to solve a problem was: 17, 20, 24, 26, 27, 30, 38, 34, 40, 35, 47, 41, 44, 49, 45, 48, 44, 54, 50, 60, 58, 58, 63, 55, 20. A. What is the type of graph you will use to represent the data? Why? 01 01 B. Give the details of the calculations and the draw the graph for above data. 02 02 C. Find sample mean and sample variance of the data. 02 02	10
Q2	What is the purpose of smoothing techniques? List different smoothing techniques. Explain any one smoothing technique in detail. 02 06 OR What are the different types of Measuring Trends methods? Explain any one method in detail with its advantages and disadvantages. 04 02 02	10
Q3	Consider a simple time series dataset where we observe daily stock prices for 10 days: $X = \{121, 123, 118, 119, 120, 124, 125, 126, 127, 129\}$ Draw the ACF plot at lag 1 and lag 2. 02 06 09	10

Q1. A Graph type : Histogram or Stem-leaf plot
reason: represent the freqⁿ distributⁿ and show spread and shape of the data. or Box plot is also accepted.



$$\text{Mean} = \frac{\sum x}{n} = \frac{1027}{25} = 41.08 \text{ sec}$$

$$\text{Variance } (s^2) = \frac{\sum (x - \bar{x})^2}{n-1} \approx \frac{183.54}{24} = 4405.12$$

Q3.

$$\rho_k = \frac{\sum_{t=k+1}^n (x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2} = -0.2 \text{ M}$$

$$\bar{x} = 123.2 - 0.1 \text{ M}$$

$$\sum (x_t - \bar{x})^2 = 111.6 - 0.1 \text{ M}$$

$$\text{lag } 0(r_0) = 1$$

$$\text{lag } 1(r_1) = \frac{74.8}{111.6} = 0.670 \quad 0.2 \text{ M}$$

$$\text{lag } 2(r_2) = \frac{34/45.12}{111.6} = \frac{0.305}{0.317} \quad 0.2 \text{ M}$$

